

D0 High-Level Design

Project: HepatoFusion

Team members: Shivam Sinay Kharangate, Nipun Chandra

Partner: Cincinnati Children's Hospital

Document: Design D0 (Week 5). D1 and D2 will sit beside this file.

AI use. An AI tool helped draft this document and the diagrams. The component breakdown, the interface definitions, and the pattern justification are the team's decisions: a file pipeline on approved shared storage, collection before integration, DICOM and Excel as the known formats, joint ownership until modality tasks are assigned, PathPresenter as the first clinical viewer for the finished pathology model, and rejection of microservices and a cloud client-server app. We can explain those choices. The tool did not choose them.

Team decisions used in this document

These are the design choices the team made before drafting D0. Later sections follow them.

Topic	Team decision
Architecture pattern	Pipeline. Each stage hands a case folder to the next.
What we start now	Radiology intake, genomics intake, and outcomes cleanup. Collection one modality at a time.
What the diagram still shows	Full path through modality integration and a research risk score, so the infrastructure has a place to land.
Transport	Files in an approved Cincinnati Children's or UC shared folder. One case folder per study ID. No public API. No consumer cloud.
Radiology format	DICOM. Conversion after DICOM is not chosen yet and needs research, so no converter is drawn.
Molecular format	Excel (.xlsx). Other molecular file types are not finalized and are blocked rather than guessed.
Missing radiology or molecular data	Working assumption: still score from the inputs that are linked, and list the missing ones as not linked. Not finalized.
Missing model-ready outcomes	Do not score.
Ownership	Both teammates on every component for now. Modality tasks will be split later when collection is further along.
Existing pathology U-Net	Already built. Not rebuilt in D0. Treated as an external system. First integration target is PathPresenter. QuPath is the other viewer under consideration.
Deployment	Not designed in D0. Spring 2027 plan.
Rejected patterns	Microservices (too many moving parts, no deployment yet) and a cloud client-server app (would send hospital files through a consumer API).

1. Title, goal statement, and conventions

Title: HepatoFusion

Goal: Put an infrastructure in place for radiology, molecular data, and cleaned clinical outcomes on the existing pediatric liver-cancer cohort, then integrate those modalities with the finished pathology localization so a pathologist can use a research risk score as a helper. The finished pathology U-Net is also planned to surface inside PathPresenter (QuPath is the alternative viewer still under consideration), so the localization is available in the systems pathologists already use.

Basic input and output. The basic input is one existing cohort case plus the new files collected for it: a DICOM CT or MRI, a molecular Excel file, and the clinical-outcomes spreadsheet. The basic output is a helper package for the pathologist: the tumor localization that the finished pipeline already produced, delivered first through PathPresenter where that integration is available, and a risk score only after integration runs, listing the inputs that were actually used.

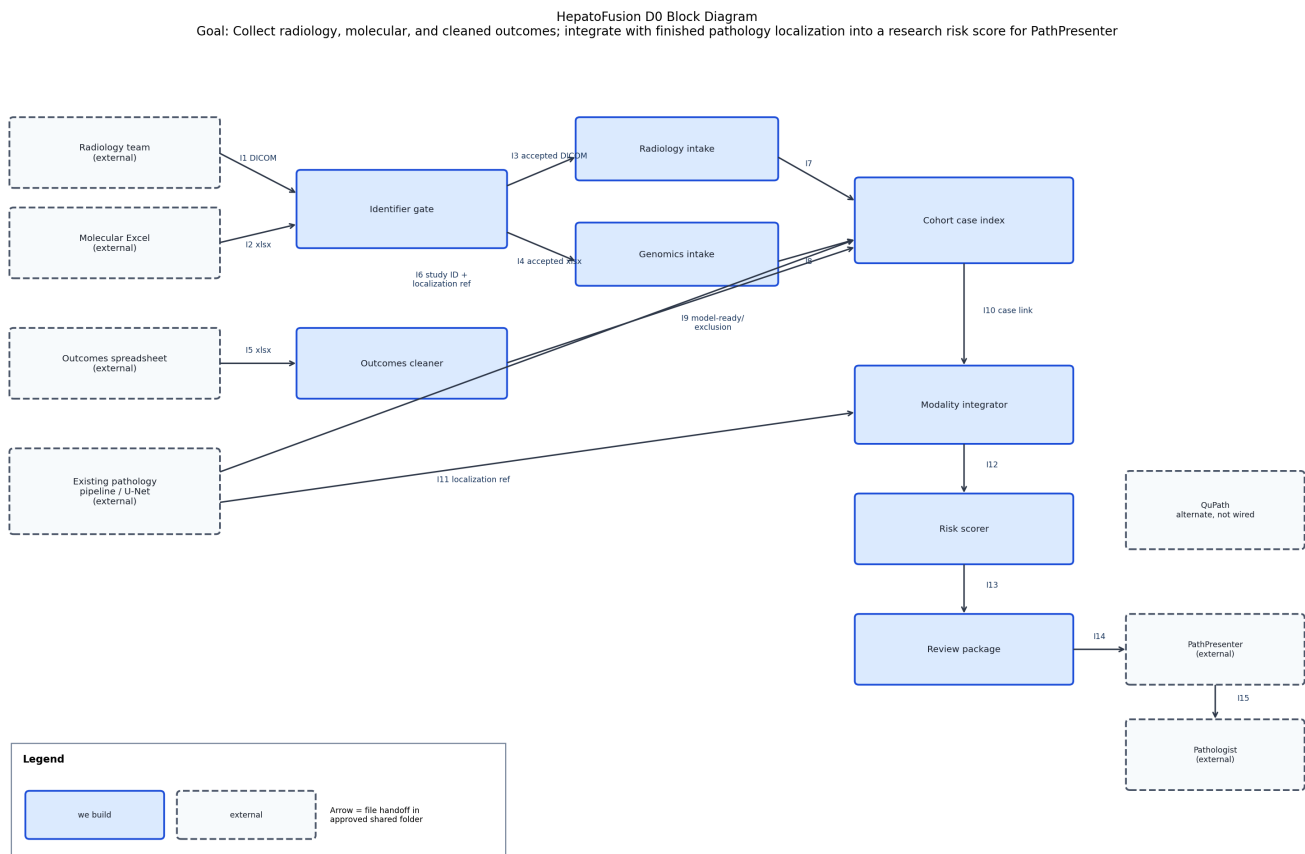
The Silver/Gold whole-slide pipeline and its U-Net are already built (1,566+ Silver and 1,546+ Gold annotations, 248+ slides, 102+ patients). D0 does not rebuild them. They appear as an external system. What we are starting now is data collection, one modality at a time. Integration and the risk score are later stages of the same pipeline, drawn here so the infrastructure has a place to land. The first planned software landing for the existing pathology model is PathPresenter. QuPath remains a second option if PathPresenter does not fit. Other modalities may later show up in those tools as well; that is not required for the collection work we are doing first. Deployment is not in this diagram. It is a Spring 2027 plan.

Conventions (also drawn on both diagrams below):

- A solid box is a component this team will build.
- A dashed box is an external system or person we depend on and will not build.
- An arrow is a file handed from one box to the next inside an approved Cincinnati Children's or University of Cincinnati shared folder. There is no public API and no consumer cloud.
- The label on an arrow (I1, I2, ...) is the interface ID in section 4.
- One case folder is one study ID. The study ID is the internal cohort key, not a patient name.

2. Block diagram

Editable source: **D0_diagrams.drawio** (sheet D0 Block Diagram), openable in diagrams.net.



Components we build (8): Identifier gate, Radiology intake, Genomics intake, Outcomes cleaner, Cohort case index, Modality integrator, Risk scorer, Review package.

External (dashed): Radiology team, Molecular Excel source, Outcomes spreadsheet, Existing pathology pipeline, PathPresenter (primary clinical viewer), QuPath (alternate viewer, not wired yet), Pathologist.

3. Component responsibility table

Owners are joint for now. Once collection is further along, each teammate will take modality tasks. That split is not decided in D0.

Component	Responsibility	In	Out	Primary owner	Story
Identifier gate	Rejects a newly collected radiology or molecular file when a direct patient identifier is present.	I1, I2	I3, I4	Shivam Sinay Kharangate and Nipun Chandra	US-05
Radiology intake	Stores an accepted DICOM study and records whether it links to one cohort case.	I3	I7	Shivam Sinay Kharangate and Nipun Chandra	US-03
Genomics intake	Stores an accepted molecular Excel and records whether it links to one cohort case.	I4	I8	Shivam Sinay Kharangate and Nipun Chandra	US-02
Outcomes cleaner	Keeps a spreadsheet row only when it joins to exactly one existing pathology case and its agreed fields are ready.	I5	I9	Shivam Sinay Kharangate and Nipun Chandra	US-04
Cohort case index	Records, for one study ID, which new files are linked to the existing pathology case.	I6, I7, I8, I9	I10	Shivam Sinay Kharangate and Nipun Chandra	US-03, US-04
Modality integrator	Builds the list of inputs actually linked for one case, including the existing pathology localization.	I10, I11	I12	Shivam Sinay Kharangate and Nipun Chandra	US-02
Risk scorer	Computes one risk score from that input list.	I12	I13	Shivam Sinay Kharangate and Nipun Chandra	US-02
Review package	Packages the existing tumor map and the score result for PathPresenter so a pathologist can use them as a research helper.	I13	I14	Shivam Sinay Kharangate and Nipun Chandra	US-01

4. Interface specification table

Protocol for every row is the same: a file written into that case's folder on approved Cincinnati Children's or UC shared storage, except I15, which is a pathologist reading the case inside PathPresenter after I14 has placed the helper files where PathPresenter can open them. Nothing in this table is a consumer-cloud API. Study IDs stand in for patient names.

ID	From → to	Inputs	Outputs	Format	Protocol	Error / handler
I1	Radiology team → Identifier gate	DICOM study, study ID	Pass or block record	DICOM + JSON status	Approved shared folder, incoming/	Non-DICOM or direct identifier present → leave incoming/, do not copy onward. Status names field type, not value. Gate handles.
I2	Molecular Excel → Identifier gate	Molecular workbook, study ID	Pass or block record	.xlsx + JSON status	Approved shared folder, incoming/	Non-Excel blocked. Other molecular types not finalized → blocked. Identifiers as in I1. Gate handles.
I3	Identifier gate → Radiology intake	Accepted DICOM, study ID, pass	Linked or unlink record	DICOM in radiology/, JSON link status	Approved shared folder, case folder	Unmatched study ID → status unmatched, no guess. No DICOM conversion yet. Intake handles.
I4	Identifier gate → Genomics intake	Accepted workbook, study ID, pass	Linked or unlink record	.xlsx in molecular/, JSON link status	Approved shared folder, case folder	Same unmatched rule as I3. Non-Excel never arrives (blocked at I2). Intake handles.

ID	From → to	Inputs	Outputs	Format	Protocol	Error / handler
I5	Outcomes spreadsheet → Outcomes cleaner	Source workbook	Model-ready row or exclusion	.xlsx in; CSV row out	Approved shared folder	Unmatched / ambiguous join / uncoded field → exclude; log study ID + field name, not patient name. Cleaner handles.
I6	Existing pathology pipeline → Cohort case index	Study ID, localization pointer	Cohort membership record	JSON pointer (file stays in place)	Approved shared folder, read-only	Study ID not in 102+ cohort → in_cohort false; no new patient. Index handles.
I7	Radiology intake → Cohort case index	DICOM link status	Updated case link record	JSON linked / not_linked / unmatched	Approved shared folder	Unmatched from I3 → radiology not linked. Index handles.
I8	Genomics intake → Cohort case index	Molecular link status	Updated case link record	JSON linked / not_linked / unmatched	Approved shared folder	Same as I7 for molecular slot. Index handles.

ID	From → to	Inputs	Outputs	Format	Protocol	Error / handler
I9	Outcomes cleaner → Cohort case index	Model-ready row or exclusion	Updated case link record	CSV row or JSON exclusion	Approved shared folder	Excluded row → outcomes not_model_ready + reason. Index handles.
I10	Cohort case index → Modality integrator	Case link record	Same record when integration runs	JSON	Approved shared folder	Not in cohort → integrator does not run. Integrator handles.
I11	Existing pathology pipeline → Modality integrator	Finished localization pointer	Localization marked present	JSON pointer	Approved shared folder, read-only	Pointer unresolved → pathology localization missing; do not rebuild U-Net. Integrator handles.

ID	From → to	Inputs	Outputs	Format	Protocol	Error / handler
I12	Modality integrator → Risk scorer	Input list for one study ID	See example payload	JSON	Approved shared folder	Outcomes not ready → scorer must not score (I13). Scorer handles refusal.
I13	Risk scorer → Review package	Input list from I12	One score, or block + no score	JSON	Approved shared folder	Outcomes not ready → risk_score null. Missing radiology/molecular: working assumption = score linked inputs only (not finalized). Scorer handles.
I14	Review package → PathPresenter	Score + localization pointer	Helper files PathPresenter can open	JSON + localization export	Approved shared folder, then PathPresenter open	Viewer open failed → leave files, record viewer_open failed; no auto QuPath fallback. Disclaimer always research prototype. Review package handles.
I15	PathPresenter → Pathologist	Case open with helper files	Pathologist review of map and optional score	PathPresenter case view	Approved hospital/institutional access	If I13 blocked score → show map + block reason, no score number. Pathologist owns clinical read.

Example payload (I12)

A case where molecular data and cleaned outcomes are in, and radiology has not been collected yet. No patient name or medical record number.

```
{
  "study_id": "HF-0102",
  "inputs": [
    {"name": "pathology_localization", "status": "linked", "source": "existing_pipeline"},
    {"name": "radiology", "status": "not_linked", "format": "DICOM"},
    {"name": "molecular", "status": "linked", "format": "xlsx"},
  ]
}
```

```
    {"name": "clinical_outcomes", "status": "model_ready"}  
  ],  
  "viewer_target": "PathPresenter"  
}
```

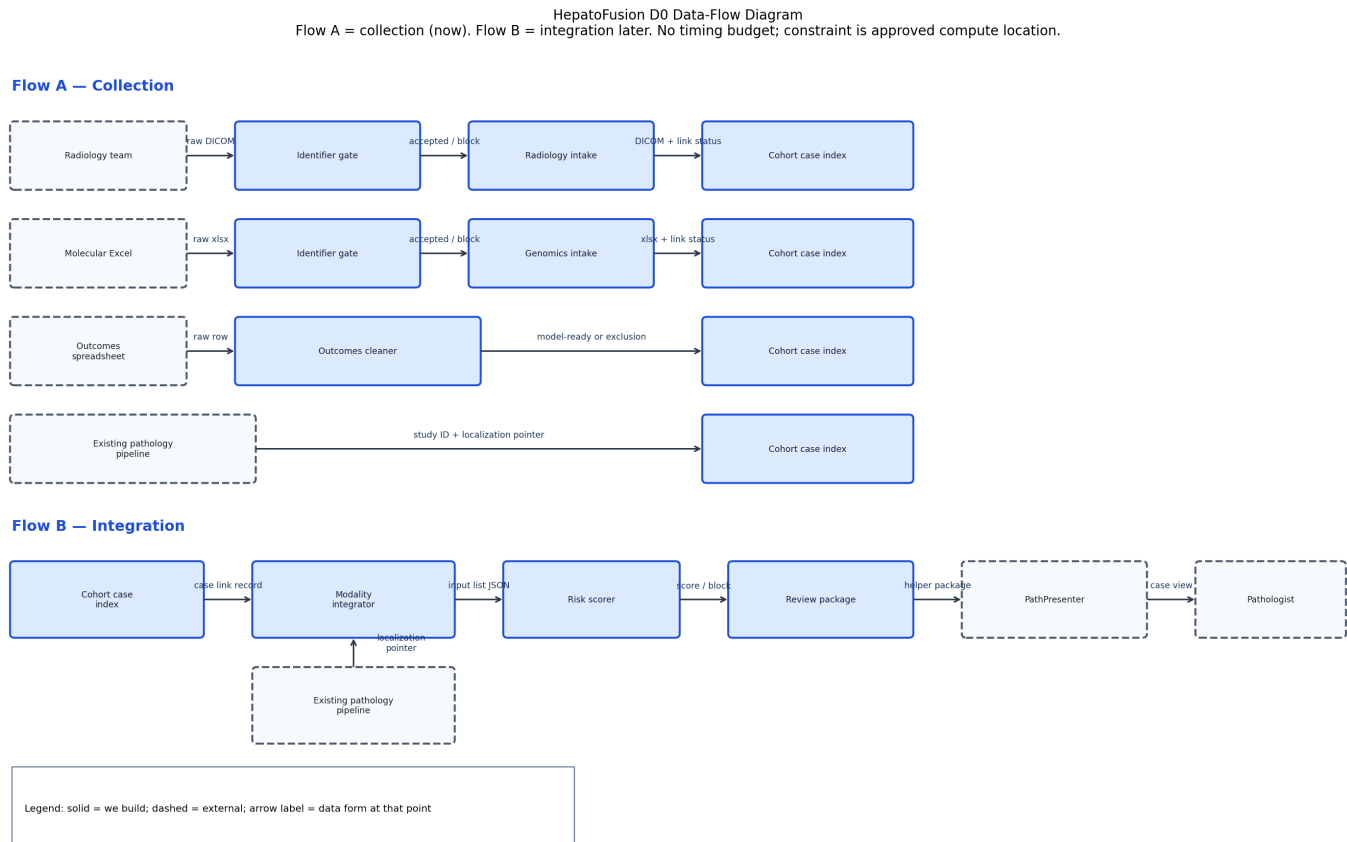
Under the working assumption, a later I13 result for this payload would name pathology, molecular, and clinical outcomes as used, would mark radiology as not linked, and would include the prototype disclaimer. That partial-score rule is not finalized. The localization itself is still intended to open in PathPresenter for US-01, even when radiology is not linked.

5. Data-flow diagram

Two flows. They share the case folder, and they are not run as one step. Collection happens per modality now. Integration runs later, when we choose to score a case. The finished pathology model also has a path into PathPresenter so the pathologist can use the localization as a helper.

No user story sets a timing budget. There is no deadline such as a maximum seconds-per-case. The constraint on these flows is where the files live (approved compute, not a personal laptop or the course repository), not a latency number.

Editable source: [D0_diagrams.drawio](#) (sheet D0 Data-Flow Diagram), openable in [diagrams.net](#).



6. Architecture pattern and justification

Pattern we are using. A pipeline. Each stage writes a file into the case folder and the next stage reads that file. The same pattern covers the whole system we will build. Flow A is the collection pipeline (gate, then intake or cleaner, then the case index). Flow B is the later pipeline (index, then integrator, then scorer, then review package into PathPresenter). We are not running them as separate products.

Fit to the problem. New files arrive one modality at a time, and the finished pathology result is already sitting there. A pipeline lets us store radiology first, molecular next, and outcomes as the spreadsheet is cleaned, then integrate only the slots that are actually linked. PathPresenter is the place the finished U-Net is meant to show up for the pathologist. That matches a research project that is collecting modalities now and integrating into an existing clinical viewer later, rather than building a new front end.

Team skills. The team is two people. Shivam's work has been the pathology and clinical side. Nipun's work has been data and systems. A file pipeline can be split later by modality without standing up a service per file type. Owners stay joint until that split is real.

Performance and timing. User stories set no response-time number. Collection is a batch of files, not a live request. The heavy model, when we train the risk score, stays on approved UC or Cincinnati Children's compute, the same constraint

as the existing U-Net work. A pipeline does not add a network hop that we would then have to budget.

Scalability. The cohort is the existing 102+ patients, not an open multi-user service. A case folder per study ID is enough for that size. We do not need independent scaling of each modality.

Hardware constraints. Week 3 constraints still hold: protected health information stays off personal machines, off the course Git repository, and off consumer cloud APIs. Training and file storage stay on approved institutional or hospital machines. There is no purchased commercial platform beyond tools already in the hospital workflow, such as PathPresenter. A folder pipeline on that storage fits. Building our own cloud-hosted viewer does not. Looking at broader deployment is a Spring 2027 plan.

Rejected. Microservices, with one service per modality. That is more moving parts than a two-person research project can operate, and there is no deployment environment to put those services in yet. Also rejected: a cloud client-server app that a pathologist would log into. That pattern would send slides, DICOM, or outcomes through a consumer API, which the hospital data-use rules and the Week 3 privacy constraint do not allow, and it would rebuild a viewer when PathPresenter is already the first integration target. We did not reject those patterns forever. We rejected them for D0 and for this academic year of research infrastructure.

7. Decision log

ID	Decision	Alternatives considered	Why this one won
D-01	The system we build is a file pipeline on approved shared storage.	Microservices, one service per modality. A cloud client-server app.	Collection is one modality after another, then a later integration step. Two people cannot operate a service per modality, and deployment is a Spring 2027 question, not this semester's. A cloud client would move hospital files onto a consumer API, which our constraints forbid.
D-02	D0 draws the full path from collection through a risk score, but the work we start now is collection and cleanup only.	Draw only the three starting pieces. Draw the risk score as if it were being built first.	The infrastructure has to have a place for files to land and a later path into risk stratification. Starting with the score would skip the data collection the radiology conversation and the spreadsheet cleanup actually require.
D-03	Components hand off files in one case folder per study ID. No HTTP API in D0.	An internal HTTP service. A database on approved compute.	The near-term work is collecting and cleaning files. A service or a database is extra structure before the formats are settled.
D-04	Radiology is stored as DICOM. No conversion stage is drawn.	Convert DICOM to another format now. Leave the radiology format unmarked.	DICOM is the form we expect from the radiology team. What happens after that, including any conversion, still needs research, so drawing a converter would pretend we had chosen one.
D-05	Molecular intake accepts an Excel workbook only.	Accept variant files or other molecular formats in the same component.	The file we know about is Excel. Other molecular forms are not finalized. Unknown forms are blocked at the gate instead of stored under a guessed type.
D-06	A case with no linked radiology or molecular file can still be scored from the inputs that are linked (working assumption, not final). A case whose outcomes row is not model-ready is not scored.	Refuse any score until radiology and molecular data are both present.	Collection will not cover every case at once, and US-02 says a missing modality must not be treated as present. The outcomes block is already the rule in UC-01.
D-07	Every component is owned by both teammates until modality tasks are assigned.	Split owners now, with Shivam on the model side and Nipun on intake.	The split will follow the modalities once collection is further along. Naming owners now would freeze a task split we have not made.
D-08	The finished pathology U-Net integrates first into PathPresenter. QuPath stays the alternate viewer and is not wired in D0.	Build a custom viewer. Wire QuPath first. Wire both PathPresenter and QuPath now.	PathPresenter is the hospital-facing system we are aiming at first for the existing localization. QuPath remains available if PathPresenter does not fit. Wiring both now would invent two viewer interfaces before the first one is proven.